

Framework for system design

Step 1 - Understand the problem and Establish design scope.

eg) Is this mobile app or web page? or both?
What are the most important features?
How many users? DAU?

Step 2: Functional Reqs / Non-Functional Reqs

- what to build?
- what is about?
- what do we do!

↳ **Quality** and scaling

- CAP Theorem
- Performance vs Scalability
- Latency vs Throughput
- Availability vs Consistency

Step 3: Back-Of-The-Envelope

Step 4: Brief design

① Core Entity

② APIs

Step 5: Data Flow (client $\leftrightarrow$ DB)
⇒ Propose high-level design and get buy-in
client → web server → service → DB
API

Step 6: Design Deep Dive with
Non-functional Reqs.

- cache
- message queue
- Rate limiter
- LB

Step 7: Wrap up